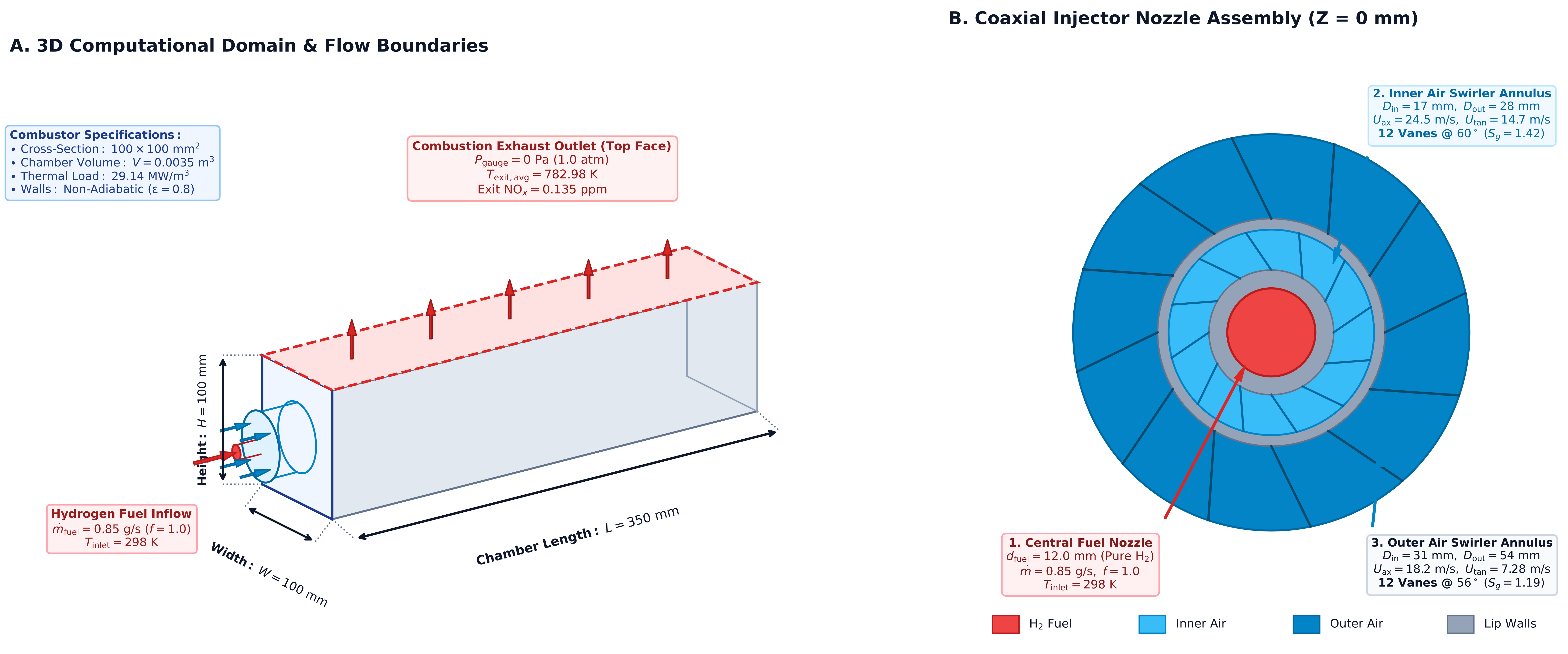
A. 3D Computational Domain & Flow Boundaries
Combustor Specifications:
• Cross-Section: 100×100 mm^2
• Chamber Volume: $V = 0.0035$ m^3
• Thermal Load: 29.14 MW/m^3
• Walls: Non-Adiabatic (ε = 0.8)
Combustion Exhaust Outlet (Top Face)
$P_{gauge} = 0$ Pa (1.0 atm)
$T_{exit,avg} = 782.98$ K
Exit $NO_x = 0.135$ ppm
Height: $H = 100$ mm
Hydrogen Fuel Inflow
$\dot{m}_{fuel} = 0.85$ g/s ($f = 1.0$)
$T_{inlet} = 298$ K
Width: $W = 100$ mm
Chamber Length: $L = 350$ mm
B. Coaxial Injector Nozzle Assembly (Z = 0 mm)
2. Inner Air Swirler Annulus
$D_{in} = 17$ mm, $D_{out} = 28$ mm
$U_{ax} = 24.5$ m/s, $U_{tan} = 14.7$ m/s
12 Vanes @ 60° ($S_g = 1.42$)
1. Central Fuel Nozzle
$d_{fuel} = 12.0$ mm (Pure H_2)
$\dot{m} = 0.85$ g/s, $f = 1.0$
$T_{inlet} = 298$ K
3. Outer Air Swirler Annulus
$D_{in} = 31$ mm, $D_{out} = 54$ mm
$U_{ax} = 18.2$ m/s, $U_{tan} = 7.28$ m/s
12 Vanes @ 56° ($S_g = 1.19$)
H_2 Fuel
Inner Air
Outer Air
Lip Walls